

Anatomy of the Truth Axis in Qwen2.5-1.5B

A working note: how attention and the FFN build, carry, and erode a one-dimensional truth signal

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Abstract

This is a *working note*, not a finished paper. It documents an exploratory, mechanistic follow-up to “How Much of Truth Fits on a Single Axis?”, in which a one-dimensional, knowledge-dependent truth representation was identified in Qwen2.5-1.5B and stress-tested with seven falsification attempts. Here I do not ask *whether* truth is linearly readable (it is), but *how the residual stream comes to carry it*: which architectural component—attention or the feed-forward network (FFN)—produces, holds, or degrades the signal, layer by layer. Three observations emerge, each with explicit caveats. (1) Decomposing the residual update into its attention and FFN contributions and fitting the unsupervised truth axis on each shows a *stable* per-layer alternation in readability (5 cross-validation seeds, two datasets). (2) Reading the axis inside the FFN’s expanded activation (post-SwiGLU, dimension 8960) does *not* improve separability over the compressed residual ($\sim +0.02$ AUC), a negative result with a clear implication for sparse-dictionary work. (3) Causal ablation overturns the intuitive “attention builds, FFN consolidates” story: attention is causally necessary in the middle band, but the FFN *degrades* the signal in the transition band (removing it *raises* truth AUC), and by the late layers the signal has already evaporated. All results are preliminary, hold for a single small model, and are not validated to the standard of the companion paper.

1 Scope and honesty statement

The companion paper reports a result that survived seven independent falsification attempts. **Nothing in this note has that status.** These are fresh, exploratory measurements on one model. Margins are sometimes modest; several findings rest on a single dataset; the behavioral label of factual knowledge is noisy because the model knows few of the probed facts. The note is written to be useful to a reader following the repository, and to fix the *method* and the *questions* for later, larger-scale work—not to claim settled conclusions. Where a statement is interpretation rather than measurement, it is marked as such.

2 Setup

The model is Qwen2.5-1.5B (base), 28 transformer blocks (layers 0–27), residual width $d_{\text{model}} = 1536$, FFN intermediate width $d_{\text{inter}} = 8960$. The truth axis is the top right-singular vector of the matrix of paired true/false residual differences, fit *without* polarity supervision, exactly as in the companion paper; separability is the held-out ROC–AUC of the projection onto that axis, evaluated with paired k -fold cross-validation. Unless noted, the dataset is NeelNanda/counterfact-tracing

(revision c945b082ca08d0a8f3ba227fb78404a09614c36e, 21,919 rows), 250 pairs; a small built-in factual set (45 pairs) is used as a clean-knowledge control. All anatomy runs use `float32`: the decomposition identity below fails spuriously in `bfloat16` because subtracting large residual states to form small per-block deltas suffers catastrophic cancellation.

3 Decomposition and its correctness gate

Within a block the residual stream is updated additively,

$$h_{\ell+1} = h_{\ell} + a_{\ell} + f_{\ell}, \quad (1)$$

where a_{ℓ} is the attention contribution (the output of `self_attn`, post-projection) and f_{ℓ} is the FFN contribution (the output of `mlp`), both captured by forward hooks. The residual stream we always studied is the *sum*; the anatomy reads the two addends separately.

Every anatomy tool embeds an **identity check**: it verifies that $a_{\ell} + f_{\ell}$ reconstructs the residual delta $h_{\ell+1} - h_{\ell}$. The verdict uses the *median* relative error across (sentence, layer), not the maximum: where a block adds almost nothing, the delta is near zero and the ratio explodes, so the maximum is fragile while the median reports the typical case. Observed median error: $\sim 1.0 \times 10^{-7}$ on both datasets—the decomposition is exact, and the numbers below are trustworthy. (The single inflated layer, block 27, has a near-zero residual delta and is excluded from interpretation.)

4 Per-layer readability: a stable alternation

For each layer I fit the truth axis on a_{ℓ} and on f_{ℓ} separately and measure held-out AUC, repeating over 5 cross-validation seeds. A per-layer difference $\text{AUC}(a_{\ell}) - \text{AUC}(f_{\ell})$ is counted as real only if its magnitude exceeds both 2σ (its own across-seed standard deviation) and a floor of 0.03. Table 1 shows the peak block.

Table 1: Truth AUC by component at the readability peak (block 15), CounterFact, mean $\pm$ std over 5 seeds. Residual = the summed stream (reference).

	residual	attention	FFN
AUC	0.792 ± 0.003	0.845 ± 0.003	0.734 ± 0.004

The alternation survives re-seeding and reproduces structurally across both datasets. On CounterFact, attention stably dominates readability at layers {6, 7, 10, 12, 13, 14, 15, 17, 18} and the FFN at {8, 11, 16, 19, 20, 21, 22, 23, 25, 26}. At the peak (block 15), where the companion paper reads truth, attention leads by $+0.110 \pm 0.004$. **Interpretation (not yet causal)**: attention appears to carry the truth signal most strongly through the middle band, including the peak. Readability, however, is not necessity—which the next section tests directly, and which changes the picture.

5 The expanded FFN space does not help

The FFN expands its input to 8960 dimensions, applies SwiGLU, and re-compresses it. One might expect the truth signal to be more cleanly readable inside that larger, putatively sparser space, before compression. I capture the expanded activation (the input to `down_proj`) and fit the truth axis there. In this $p \gg n$ regime the permutation null is elevated, so only the margin over the

null counts: at the peak the expanded axis clears its null cleanly (built-in 0.953 vs. null 95th-pct 0.669, margin +0.284, $p = 0.0099$; CounterFact 0.816 vs. 0.561, margin +0.254, $p = 0.0099$). **But relative to the compressed residual at the same layer, the expanded space gains essentially nothing:** +0.017 (built-in) and +0.023 (CounterFact). Raw architectural expansion does not undo superposition on these facts. The implication for future work is concrete: sparsity must be *imposed* (e.g. a trained sparse autoencoder), not merely afforded by a wider space.

6 Causal ablation: attention builds, the FFN erodes

To separate readability from necessity I zero a component’s contribution over a band of layers (a hook returns zeros, so $h + 0 = h$), let the effect propagate, and measure the collapse of truth AUC at a downstream readout (Table 2).

Table 2: Causal ablation on CounterFact (250 pairs). Drop = intact – ablated; positive drop = the component is causally necessary.

band (layers)	readout	intact	attention-off (drop)	FFN-off (drop)
11–15 (middle)	block 15	0.793	0.487 (+0.306)	0.819 (−0.026)
16–18 (transition)	block 18	0.712	0.708 (+0.004)	0.815 (−0.103)
19–26 (late)	block 27	0.537	0.534 (+0.003)	0.539 (−0.002)

Three readings, in order of confidence:

- **Middle band: attention is causally necessary.** Removing attention collapses truth from 0.793 to 0.487 (drop +0.306); removing the FFN does nothing (−0.026). Attention does not merely read the signal here—it builds it.
- **Transition band: the FFN *degrades* the signal.** Removing attention again does nothing (+0.004), but removing the FFN *raises* truth AUC from 0.712 to 0.815 (drop −0.103). From layer 16 onward the FFN injects content that lowers true/false separability. This is the first causal support for an earlier conjecture—that the FFN adds context which amplifies the overlap on uncertain facts—though it rests on one band and one dataset and should be treated as a lead, not a result.
- **Late band: nothing remains to destroy.** The intact AUC is already ~ 0.54 (near chance): the truth signal has evaporated toward the output, so neither ablation has an effect.

This falsifies the initial “attention builds, FFN consolidates” hypothesis. There is no late consolidation; the FFN’s apparent late *readability* dominance (Section 5) was the readability of a dying signal, not active maintenance. The corrected picture: attention constructs the truth representation in the middle layers; the FFN begins to erode it in the transition band; by the late layers it is gone, as the model shifts from representing the fact to predicting the next token.

7 Limitations

One small model; results may not transport to larger models that know more facts and may organize truth differently—verifying the phenomenon (not just the method) on a larger model is the natural next step. Margins in the expanded-space and transition-band analyses are modest and

based on single datasets. The factual knowledge label is noisy. Ablation measures necessity for *linear readability* of truth, not for the model’s behavior. None of these findings has undergone the falsification battery of the companion paper; they are documented here as a working record and a set of sharpened questions for sparse-dictionary and porting work.

Reproducibility

Tools (`anatomy.py`, `anatomy Consolidate.py`, `anatomy Expanded.py`, `ablation.py`) reuse the companion repository’s data and geometry code; each embeds the identity / null / stability shields described above. Dataset revision is pinned in Section 2. License: CC BY 4.0.